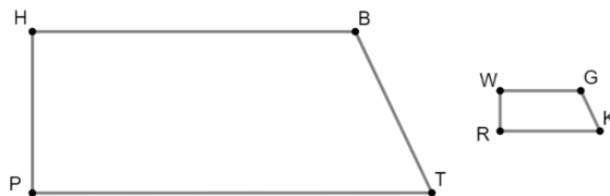


Geometry
Unit 6
Lesson 13 Practice

Name _____
Date _____
Period _____

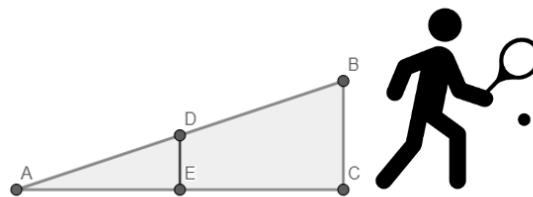
Wendy, a party planner, is designing a banner for a retirement party. Her client wants two similar banners: a large banner, $HBTP$, for the main entrance, and a similar sized banner, $WGKR$, for a photo booth.

Wendy knows that $PT = 72$ inches (in),
 $HP = (5x - 3)$ in., $WG = 6$ in., $WR = (3x - 10)$ in.,
 $BT = 40$ in., and $RK = 9$ in.



1. What scale factor will map $WGKR$ to $HBTP$?
2. What is the perimeter of $HBTP$, the main entrance banner?
3. What is the perimeter of $WGKR$, the photo booth banner?

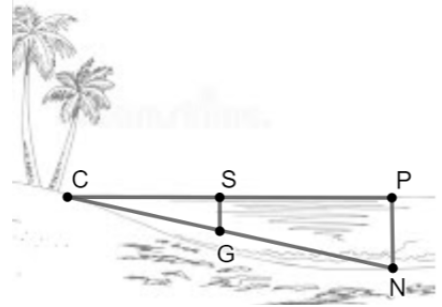
In the diagram, $\triangle ADE \sim \triangle ABC$, $AE = 7$ meters (m),
 $DE = 0.93$ m, and $EC = 21$ m. Serena is doing a
tennis drill where she wants to hit the ball so that
it will just pass over the net, $\overline{DE}$, and land 7 m
away from the base of the net.



4. What scale factor will map $\triangle ABC$ to $\triangle ADE$?
5. What is the height, represented by $\overline{BC}$, of the tennis racket?

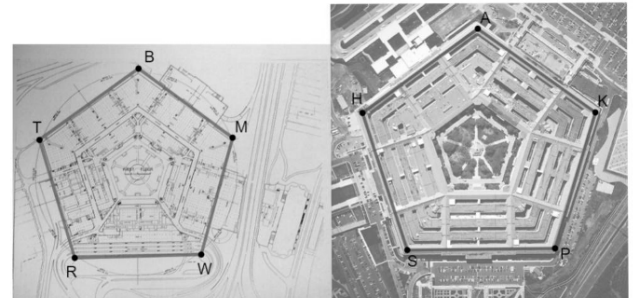
A lake front property owner is building a dock which extends the length of $\overline{CP}$ into the lake. $\overline{PN}$ and $\overline{SP}$ are the two main support posts for the dock.

$\triangle CSG \sim \triangle CPN$, $CS = 6.5$ feet (ft), $SG = (2x - 1)$ ft,
 $PN = (3x + 9)$ ft, and $CP = 32.5$ ft.



6. What scale factor will map $\triangle CPN$ to $\triangle CSG$?
7. What is the depth of the longer support post, represented by $\overline{PN}$?
8. What is the depth of the shorter support post, represented by $\overline{SG}$?

Cesar wants to create a scale model of the Pentagon, where all sides are equal, for his social studies project. A copy of the blueprint, $BMWRT$, will guide him in building a similar scale model, $AKPSH$. The blueprint's sides, $BMWRT$, are $\frac{1}{6}$ of the scale model Pentagon, $AKPSH$.
 $TB = (0.5x + 5)$ centimeters (cm) and
 $HA = (5x + 10)$ cm.



9. What is the perimeter of the Pentagon in the blueprint, $BMWRT$?
10. What is the perimeter of the scale model Pentagon, $AKPSH$?